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WIRELESS SECURITY SYSTEM
Using PIR Sensors

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This project demonstrates a security system in which four pyroelectric infrared (PIR) motion sensors are placed in four sides—front, back, left and right—of the area to be covered. It detects motion from any side and turns on the audio-visual alarm. It also displays the side where the motion (intruder) is detected. All sensors send signal to the central controller circuit wirelessly. The author's prototype arrangement is shown in Fig. 1.

Fig. 1: Author's prototype arrangement

System block diagram

System block diagram of the wireless security system is shown in Fig. 2. The project uses PIR motion sensors to detect motion and ASK-based radio frequency (RF) transmitter and receiver modules to send signals wirelessly. It uses AT89S52 microcontroller (MCU) and LCD to display the side where the motion is detected.

The block diagram has two parts: four transmitter units with PIR sensors, encoders and RF transmitters; and receiver unit with RF receiver, decoder, MCU and some audio-visual circuits.

Transmitter unit. There are four transmitter blocks, one each for front, back, left and right sides. Each block consists of a PIR sensor, RF encoder chip and RF transmitter (Tx) module.

PIR sensor. The PIR sensor detects motion by measuring any change in IR levels emitted by objects. Pyroelectric devices have elements made of a crystalline material, which generate an electric current when exposed to IR radiation. Changes in the amount of IR falling on such devices changes the voltages that are generated. Sensor output goes high when it detects motion. Sensor output is given to RF encoder chip.

RF Encoder. HT12E encoder chip encodes PIR sensor output into serial bit streams and gives it to RF Tx module.

RF transmitter. ASK transmitter modulates incoming digital signals from the RF encoder using a 434MHz carrier and transmits it through its antenna.

Receiver unit. This is the central controlling section that receives signals from any of the four PIR sensors at the transmitter side. It gives audio and visual alarms, and displays the side where the motion is detected on LCD1.

RF receiver. This module operates on 434MHz frequency and demodulates the received digital signals before sending bit streams to RF decoder chip.

RF decoder. HT12D RF decoder chip decodes bit streams and generates parallel 4-bit digital output, which is given to the MCU.

MCU. AT89S52 MCU performs the following tasks:

1. Detects side where motion is detected from RF decoder digital output
2. Displays various messages including side of detected motion on LCD1
3. Turns on speaker and blinks LED for audio-visual alarm when motion is detected

LCD panel. The 16x2 LCD panel displays messages given by the MCU.

Multivibrators. There are two multivibrators. One for blinking the

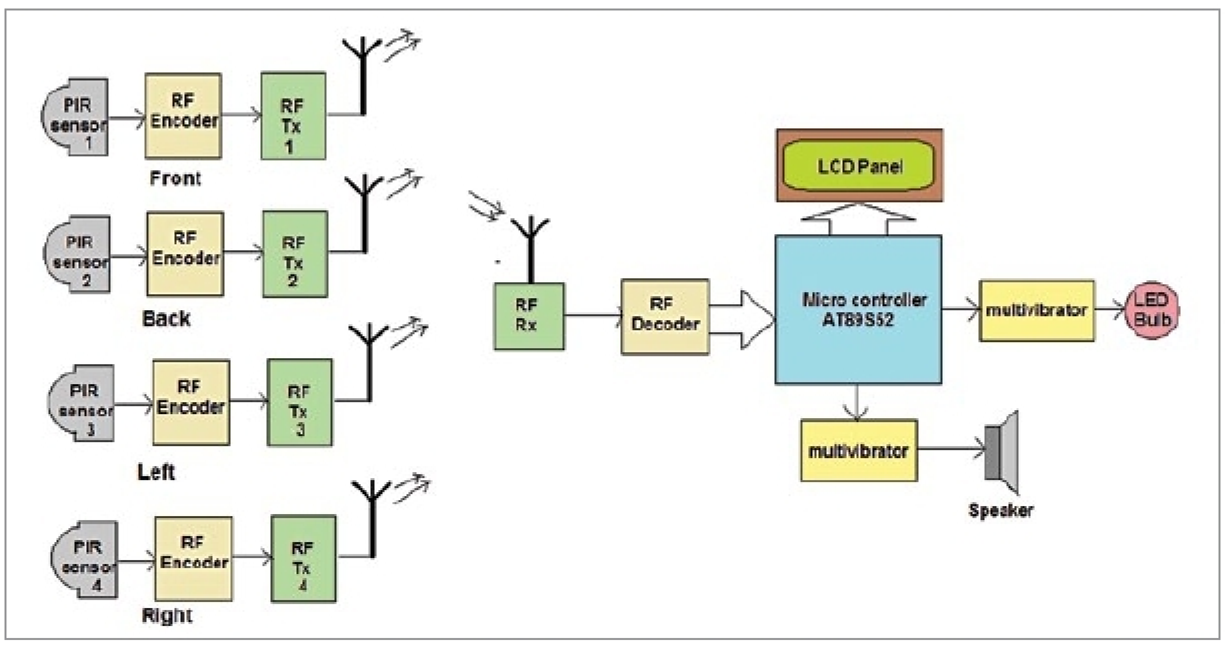

Fig. 2: Block diagram of wireless security system using PIR sensors